

Senior Secondary Two (SS2)
Agricultural Science Scheme of Work – First Term, 2026/2027

Week	Topic	Content
1	Farm Surveying and Planning	Meaning of farm surveying (important) Surveying equipment and use: i) ranging pole ii) gunter's chain iii) measuring tape iv) prismatic compass v) theodolite vi) offset vii) plumb bob viii) arrow, etc. Maintenance measures on farm survey equipment: i) wash and drying ii) oiling and greasing iii) tightening screw and nut Importance of farm surveying: i) provide farm dimension ii) for feasibility studies iii) help to determine management technique
2	Farm Planning	Definition of farm planning and the design of a farmstead: i) water supply ii) accessibility iii) soil type iv) slopes
3	Principles of farmstead planning and location of farm building/structure	Meaning and importance of farmstead planning: i) definition of goal of farm business ii) achievement of the goal Principles of farmstead layout Method of farm layout survey: i) grid survey ii) non-grid/free survey iii) others (transect, traverse)
4	Forest management	Definition of forest and forestry Importance of forest and forestry Forest regulation in Nigeria Forest management practices in Nigeria: i) forest regulation ii) selective exploitation iii) deforestation iv) regeneration v) afforestation
5	Agroforestry practices in Nigeria	Meaning and practices of agroforestry: i) taungya system Different agroforestry combinations Types of tree species suitable for agroforestry practice
6	Floriculture/Ornamental plants	Definition of floriculture Meaning and importance of ornamental plants Common types of ornamental plants: i) bedding plants (mostly flowering plants) ii) hedging plants iii) lawn plants iv) landscape plants v) pot plants, etc. Examples of ornamental plants

Week	Topic	Content
		Setting and locations: i) identification of locations/settings requiring ornamental plants (avenues/streets, building environment, gardens) Determination of ornamental plant for each location Recognition of ornamental plants using their common and botanical names
7	Floriculture/Ornamental plants (cont'd)	Methods of culturing ornamental plants: i) method of propagation by seed ii) vegetative propagation Importance Methods of propagation Maintenance operations of ornamental plants Purpose for carrying out each operation
8	Disease of crops	Meaning of disease General effect of disease on crop production Important diseases of major crops: i) cereals – smut, blast, rust, leaf curl ii) legumes – cercospora, leaf spot, rosette iii) sigatoka iv) beverages – black pod, swollen shoot, leaf rust v) tubers – mosaic, bacterial blight, etc. vi) fruits – citrus gummosis, dieback vii) fibre – black arm, bacterial leaf blight of cotton, etc. viii) vegetables – root knot of tomato/okro, damping off ix) onion twister, etc. x) stored produce – mould, etc. Causal organism of each crop disease: bacteria, fungi, nematode, etc. Description of symptoms of each disease Prevention and control measures: i) cultural methods ii) biological control iii) chemical control
9	Pest of crops	Meaning of pest Classification of pest: i) insect ii) non-insect pest Classification of pest based on mouth parts: i) biting and chewing ii) piercing and sucking iii) boring Important pests of major crops: i) cereals – stem borer ii) legume – pod borer

Week	Topic	Content
		iii) beverage – cocoa mirid iv) tuber – yam beetle, locust, cassava mealybug, etc. v) fibre – cotton stainer, etc. vi) fruits and vegetables – grasshopper, locust, etc. vii) storage pest of farm produce – weevil, bean beetle, etc. Other important pests (non-insect pest e.g. bird, rodent): i) nature of damage ii) biological description of selected insect pests General prevention and control measures: i) cultural methods ii) biological methods iii) chemical methods iv) quarantine methods Side effects of various prevention and control measures: i) pollution ii) poisoning iii) disruption of ecosystem Economic importance of pest
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NAME OF SCHOOL: Mater Misericordiae Secondary School, Rumuomasi

TERM: First Term 2026/2027 academic session

WEEK: ONE

CLASS: SS2

SUBJECT: Agricultural science

TOPIC: Farm surveying and planning

Meaning of Farm Surveying

Farm surveying is the art and science of taking measurements on a piece of farmland in order to determine its size, shape, boundaries, and features, and to represent this information accurately on a map or plan. It involves the use of specific instruments to measure distances, angles, elevations and directions on the field before laying out a farm.

Importance/Purposes of Farm Surveying

1. To determine the exact size (area) and boundaries of a farmland.
2. To provide accurate data for preparing a farm map or plan.
3. To assist in the proper layout of farm structures, roads, drainage and irrigation systems.
4. To resolve boundary disputes between neighbouring farmers/landowners.
5. To aid in the sub-division of land into plots for different farm enterprises.
6. To assist in levelling farmland for irrigation and erosion control.

Surveying Equipment and Their Uses

Equipment	Description	Use
Ranging Pole	A straight wooden or metal pole, about 2–3 m long, painted in alternate bands of red and white (or black and white) for visibility, with a pointed metal shoe at one end.	Used to mark and sight straight lines on the field, to indicate positions of survey stations, and to help the surveyor align the chain or tape correctly.
Gunter's Chain	A measuring chain made of 100 links of galvanised steel wire joined by rings, with a total length of 66 feet (22 yards), fitted with brass handles at both ends and tally tags at every 10 links.	Used for measuring distances and chaining farmland during land surveying, especially for laying out plots and farm boundaries.
Measuring Tape	A flexible tape made of steel, linen, fibre-glass or synthetic material, usually wound in a case, and marked in metres/feet.	Used for measuring shorter and more precise distances, such as the length and width of plots, buildings, and farm structures.
Prismatic Compass	A hand-held instrument with a magnetic needle, graduated circular dial and a sighting prism.	Used to determine directions (bearings) of farm boundaries and to help the surveyor orient the survey plan correctly to the North.

Equipment	Description	Use
Theodolite	A precision optical instrument mounted on a tripod, consisting of a telescope that can rotate both horizontally and vertically, with graduated circles for reading angles.	Used for measuring horizontal and vertical angles accurately, and for determining elevations and levels on farmland.
Offset (Offset Rod/Staff)	A short graduated rod, usually about 3 feet (1 m) long, pointed at one end.	Used for measuring short perpendicular distances (offsets) from the main survey (chain) line to fix the position of farm features such as fences, buildings, or trees.
Plumb Bob	A small, pointed, cone-shaped weight made of brass or lead, suspended from a string/cord.	Used to establish an exact vertical line from a point, to transfer ground points to instrument stations, and to check that ranging poles/instruments are perfectly upright.
Arrows (Marking/Chaining Arrows)	A set of thin steel rods, pointed at one end and bent into a loop or ring at the other, usually about 10 in a set.	Used to mark the end of each chain length while chaining, so that the surveyor can keep an accurate count of the number of chain lengths measured.

Maintenance Measures on Farm Surveying Equipment

Proper care and maintenance of surveying equipment prolongs their lifespan, ensures accuracy of measurement, and reduces the cost of frequent replacement. The following measures should be observed:

1. Clean equipment thoroughly after each use to remove soil, mud, dust and moisture.
2. Store equipment in a dry, cool and well-ventilated store to prevent rusting and damage by moisture.
3. Apply light oil/grease on metal parts such as chains, arrows and the shoe of ranging poles to prevent rust.
4. Handle delicate instruments such as the theodolite and prismatic compass with care to avoid mechanical damage to their lenses and graduated dials.
5. Keep the theodolite and compass in their padded cases/boxes when not in use, and protect them from dust and direct sunlight.
6. Repaint ranging poles regularly and repair any bent or broken parts.
7. Check the gunter's chain periodically for stretched, worn-out or broken links and repair or replace them promptly.
8. Avoid dropping or throwing equipment about, as this can affect the accuracy of readings.
9. Ensure equipment is properly calibrated/tested for accuracy before use.
10. Keep an inventory/record of all surveying equipment and carry out regular inspection.

Assignment:

1. Draw and label farm surveying equipments discussed in class.
2. Define farm planning.

NAME OF SCHOOL: Mater Misericordiae Secondary School, Rumuomasi

WEEK: Two

CLASS: SS2

SUBJECT: Agricultural science

TOPIC: Farm planning

: Definition of Farm Planning

Farm planning is the process of deciding in advance what farm enterprises to undertake, how to organise farm resources (land, labour, capital and management), and how to arrange farm structures and operations so as to achieve maximum production at minimum cost over a given period.

It involves the systematic arrangement of farm activities, structures and resources to ensure efficiency, convenience and profitability on the farm.

The Farmstead

A farmstead refers to the residential and operational centre of a farm — the area where farm buildings and structures such as the farmhouse, barns, stores, livestock houses, workshops and other farm structures are sited.

The design of a farmstead is the deliberate arrangement and layout of these structures on the farm to allow for easy movement, efficient operation, and proper use of farm resources.

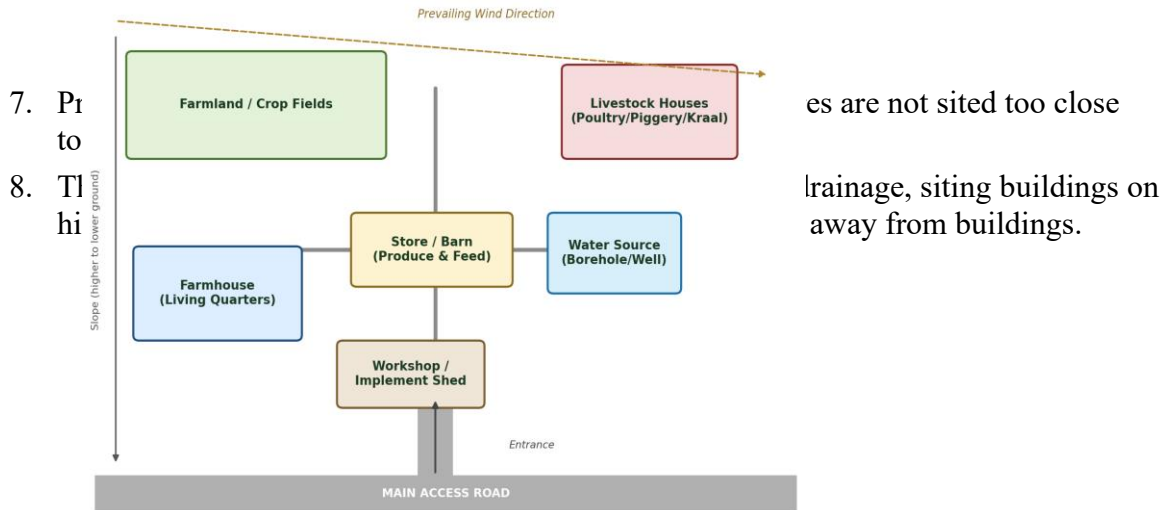
Design of a Farmstead

The design of a farmstead refers to the planned arrangement of farm buildings and structures — such as the farmhouse, livestock houses, stores, barns, workshop and processing units — on the farm in a way that promotes efficiency, convenience, hygiene and safety.

A well-designed farmstead follows certain principles:

1. The farmhouse (living quarters) should be sited close to the entrance/access road, but at a reasonable distance from livestock houses to reduce noise, odour and disease risk.
2. Livestock houses (e.g. poultry pen, piggery, cattle kraal) should be sited downwind of the farmhouse, so that odour and dust are blown away from the residential area.
3. The store and barn for keeping farm produce, feed and implements should be centrally located for easy access from the farmhouse, farmland and livestock houses.
4. A workshop or implement shed should be positioned near the farm entrance for easy movement of machinery in and out of the farm.
5. Structures should be spaced adequately apart to allow free movement, proper ventilation, and to control the spread of fire or disease.
6. Roads/paths within the farmstead should link all structures and connect to the main access road for easy movement of workers, produce and machinery.

Fig. 1: Layout Plan of a Typical Farmstead



Factors to Consider in the Design and Layout of a Farmstead

1. Water Supply

The farmstead should be sited close to a reliable and adequate source of water (stream, river, well, borehole or reservoir).

Water is needed for drinking (by farm workers and livestock), irrigation, processing of farm produce, and general farm sanitation.

A poor water source increases cost of production and reduces efficiency.

2. Accessibility

The farmstead should be located along or near a good, motorable road that is accessible in all seasons.

Good accessibility makes it easy to transport inputs (seeds, fertilizer, feed) to the farm and farm produce to the market.

It also allows easy movement of workers, farm machinery and visitors, and quick response in emergencies (e.g. veterinary or medical help).

3. Soil Type

The soil at the site should be suitable for the intended crops or, for structures, firm enough to bear the weight of buildings.

Soil should have good drainage; waterlogged or poorly drained soils are unsuitable for siting buildings as they may cause flooding, dampness and erosion.

Soil fertility and texture should also be considered where the farmstead adjoins cropland.

4. Slope/Topography of the Land

A gentle slope is preferred for a farmstead — it allows for natural drainage of water away from buildings.

A very steep slope encourages soil erosion and makes construction and movement difficult, while a completely flat, low-lying area may encourage flooding and water stagnation.

The land should also be considered in relation to wind direction, to reduce the spread of odour and disease from livestock houses to the farmhouse.

5. Nearness to the market for farm produce and inputs.
6. Availability of labour in the area.
7. Security of the farm and its produce.
8. Size and shape of the farmland.
9. Government policy and land tenure system.

Importance of Farm Planning

1. It ensures efficient use of farm resources (land, labour, capital).
2. It reduces cost of production and wastage.
3. It allows for proper arrangement of farm structures for convenience and safety.
4. It helps in achieving maximum farm output and profit.
5. It aids proper record keeping and future expansion of the farm.

Calculation of Area of Farmland

Before a farm can be planned, the area of the farmland must be known, since it determines how many crops/plants or structures the land can take. Farmland is usually measured in square metres (m²) or hectares (ha).

Note: 1 hectare (ha) = 10,000 m²

The area of a plot of farmland is calculated according to its shape:

Shape of Land	Formula for Area
Square	Area = Length × Length (side × side)
Rectangle	Area = Length × Breadth
Triangle	Area = $\frac{1}{2} \times \text{Base} \times \text{Height}$
Circle	Area = $\pi \times r^2$ ($\pi \approx 22/7$ or 3.142; r = radius)
Trapezium	Area = $\frac{1}{2} \times (\text{Sum of parallel sides}) \times \text{Height}$, i.e. $\frac{1}{2} \times (a + b) \times h$

Worked Example 1

A rectangular farmland measures 200 m long and 50 m wide. Calculate its area in hectares.

$$\text{Area} = \text{Length} \times \text{Breadth} = 200 \text{ m} \times 50 \text{ m} = 10,000 \text{ m}^2$$

$$\text{Area in hectares} = 10,000 \text{ m}^2 \div 10,000 \text{ m}^2 \text{ per ha} = 1 \text{ hectare}$$

Worked Example 1b (Trapezium)

A farmland is in the shape of a trapezium with its two parallel sides measuring 80 m and 60 m, and the perpendicular height between them measuring 40 m. Calculate the area of the land.

$$\text{Area} = \frac{1}{2} \times (a + b) \times h = \frac{1}{2} \times (80 \text{ m} + 60 \text{ m}) \times 40 \text{ m}$$

$$\text{Area} = \frac{1}{2} \times 140 \text{ m} \times 40 \text{ m} = 2,800 \text{ m}^2$$

Calculation of Plant Population

Plant population is the total number of plants (or stands) that can be raised on a given piece of land. It is important in farm planning because it helps the farmer determine the quantity of seeds/seedlings, fertilizer and other inputs needed, and helps in estimating expected yield.

Plant population is calculated using the formula:

$$\text{Plant Population} = \text{Area of Land} \div \text{Area of Land Required per Plant}$$

Where the area of land required per plant = spacing between rows $\times$ spacing between plants (within a row).

Worked Example 2

A farmer wants to plant maize on a plot of land measuring 1,000 m² at a spacing of 100 cm $\times$ 50 cm. Calculate the expected plant population.

$$\text{Area required per plant} = 100 \text{ cm} \times 50 \text{ cm} = 1 \text{ m} \times 0.5 \text{ m} = 0.5 \text{ m}^2$$

$$\text{Plant population} = \text{Area of land} \div \text{Area per plant} = 1,000 \text{ m}^2 \div 0.5 \text{ m}^2 = 2,000 \text{ plants}$$

Note: Where the farmland is measured in hectares, it should first be converted to square metres (1 ha = 10,000 m²) before applying the formula, and all spacing measurements should be converted to the same unit (metres) before multiplying.

Assignment:

1. A farmer has a triangular plot of land with a base of 60 m and a height of 40 m. He wishes to plant yam at a spacing of 1 m $\times$ 1 m. Calculate (a) the area of the land and (b) the expected plant population.

NAME OF SCHOOL: Mater Misericordiae Secondary School, Rumuomasi

WEEK: THREE

DATE: 25th September, 2026

CLASS: SS2

SUBJECT: Agricultural science

TOPIC: Principles of farmstead planning and location of farm building/structure

Meaning and Importance of Farmstead

A farmstead refers to the area of land on a farm that contains the farmhouse and other farm buildings and structures, such as barns, stores, silos, animal housing units, workshops, and processing units. It is the operational centre of the farm business from which farm activities are coordinated and managed.

Importance of a farmstead:

- i. It serves as the administrative and operational headquarters of the farm.
- ii. It provides storage for farm produce, tools, machinery, and inputs.

- iii. It houses livestock and protects them from weather and predators.
- iv. It provides accommodation for farm workers and the farm family.
- v. It facilitates easy supervision and control of farm operations.
- vi. It enhances efficient processing and value addition to farm produce.

Importance of Farmstead Planning

Farmstead planning is the deliberate arrangement and siting of farm buildings and structures in a manner that promotes efficiency, safety, and convenience in farm operations. Its importance can be considered under the following:

i) Definition of the goal of the farm business

Planning helps the farmer to clearly define the objectives of the farm business, such as the type of enterprise (crop, livestock, or mixed farming), scale of production, and expected output, so that farm buildings and structures are sited to support these objectives.

ii) Achievement of the goal

A well-planned farmstead makes it possible to achieve the set goals by ensuring that resources (land, labour, capital, and structures) are efficiently combined. It reduces cost of production, minimises time and energy wasted in movement, prevents overlapping of activities, and creates room for future expansion of the farm business.

Principles of Farmstead Layout

The following principles guide the layout of a farmstead:

1. Accessibility: Buildings should be sited close to good roads for easy movement of produce, inputs, and machinery.
2. Topography: The layout should take advantage of the natural slope of the land for good drainage.
3. Convenience and efficiency: Related buildings and structures should be sited close to one another to reduce time and labour spent moving between them.
4. Health and sanitation: Livestock houses should be sited away from the farmhouse and water sources to prevent contamination and disease.
5. Provision for expansion: Enough space should be left around existing structures for future growth of the farm business.
6. Security: The farmstead should be sited in a location that can be easily secured against theft, fire, and flooding.
7. Water supply and drainage: structures should be close to a reliable source of water, with good provision for drainage of waste water.
8. Direction of wind: Livestock and waste-disposal structures should be sited downwind of the farmhouse to reduce odour and disease spread.
9. Economic use of space: The layout should make maximum use of available land without overcrowding.

Method of Farm Layout Survey

Before a farmstead is laid out, the farm site is surveyed to determine the exact size, shape, and features of the land. The following methods are commonly used:

i) Grid Survey

This involves dividing the farm land into a network of squares or rectangles (a grid) using measuring tapes and ranging poles. Measurements are taken systematically along straight lines that cross one another at right angles, and the readings are used to plot the map of the farm. It is suitable for fairly flat and regularly shaped land.

ii) Non-grid (Free) Survey

This method does not follow a fixed grid pattern. Instead, the surveyor takes measurements based on the natural boundaries and features of the land, such as irregular boundaries, streams, and rock outcrops. It is more flexible and is often used for irregularly shaped farmland.

iii) Other Methods (Transect and Traverse)

Transect survey: A straight line is drawn across the farmland, and measurements/observations (such as soil type, vegetation, and slope) are recorded at regular intervals along the line.

Traverse survey: A series of straight lines (traverse lines) are measured and connected end to end round the boundary of the farmland, with angles recorded at each point, to produce an accurate outline of the farm.

Assignment:

1. Define Forest, forestry and silviculture.
2. List six importance of forest.

NAME OF SCHOOL: Mater Misericordiae Secondary School, Rumuomasi

WEEK: Four

DATE: 2nd October, 2026

CLASS: SS2

SUBJECT: Agricultural science

TOPIC: Forest Management

Meaning of Forest, Forestry, and Silviculture

A **forest** can be defined as a large area of land covered with trees and bushes, either growing wild or planted for some specific purposes.

Forestry is the study and management of forest and forest resources.

Silviculture is the study of forest trees and other related trees.

Importance of Forest

- 1. Provision of food:** Provides food products such as fruits, bush meat, and vegetables.
- 2. Provision of fuel:** Dead forest wood serves as firewood or a source of fuel used for cooking and other domestic purposes.
- 3. Provision of timber:** Forest provides timber for furniture and other domestic and industrial constructions.
- 4. Provision of medicinal herbs:** Provides medicinal herbs used by local healers and pharmaceutical industries.
- 5. Formation of rain:** Forests, because of their cool environment, help in the condensation of water vapour resulting in the formation of rain.
- 6. Provision of employment:** Forests provide employment for many people, ranging from forest guards, saw millers, and forest lawyers to other workers concerned with lumbering.
- 7. Prevention of erosion:** Soil erosion is reduced or even prevented in a forest due to the vegetative cover provided by the trees.
- 8. Provision of windbreak:** Forest trees, especially in Northern Nigeria, serve as windbreaks. This reduces the speed of wind, thereby controlling wind erosion and desert encroachment.
- 9. Source of foreign exchange:** Provides foreign exchange earnings for the country through the export of timber and its by-products.
- 10. Purification of atmospheric air:** Purifies air through the removal of carbon (IV) oxide and addition of oxygen to the atmosphere during photosynthesis and respiration, respectively.

Forest Management Practices in Nigeria

In order to ensure the continuous supply of forest and its products, the following management practices should be judiciously adopted:

1. Forest Regulation

These are laws promulgated by the government in the form of edicts, decrees, and bye-laws to prevent people from exploiting or indiscriminately tapping forest resources, hence preserving the forest. Forest regulations in Nigeria include:

- i. The prohibition of bush burning.
- ii. Encouraging people to plant trees.
- iii. Ban on indiscriminate cutting or felling of timber trees.

- iv. Ban on farming in forest reserves.
- v. Ban on the cutting down of underage trees.
- vi. Ban on collection of leaves and firewood from the forest.
- vii. Requirement for people to obtain licenses/permissions to enable them to cut down trees for human needs.

2. Selective Exploitation

Selective exploitation is the process of cutting only mature trees of about 20–25 years or when a particular size is reached.

Advantages of Selective Exploitation:

- i. Protects the soil from erosion.
- ii. Prevents indiscriminate felling of trees.
- iii. Ensures a continuous supply of timber.
- iv. Allows the elimination of undesirable or unwanted species of timber.

3. Deforestation

Deforestation is the continuous removal of forest trees either by bush burning or indiscriminate felling of trees, without replacing them.

Causes of Deforestation:

- i. Unfavourable climatic factors:** Persistent and prolonged drought can lead to the death of forest species.
- ii. Wrong farming activities:** Can cause forest destruction through the use of forest land for crop production or grazing livestock, and practicing bush fallowing or shifting cultivation.
- iii. Timber exploitation:** The practice of selective elimination of certain tree species in a natural forest causes deforestation.
- iv. Mining / Industrialization:** Forest trees are destroyed when land is cleared for excavation and mining of minerals.
- v. Natural disasters:** Fire is the most serious hazard to which forests are exposed, especially during the dry season (often caused by fires set by farmers). Other natural disasters include volcanic eruptions and earthquakes.

Effects of Deforestation:

- i. It encourages soil erosion.
- ii. Leads to desertification.
- iii. It reduces the amount of rainfall in an area.
- iv. Increases leaching of soil nutrients.

- v. It decreases soil moisture content.

4. Regeneration

Regeneration is the process by which young trees develop to replace old, mature trees as they are cut down. It is the process of forest regrowth after it has been exploited. There are two types of regeneration:

- a) Natural regeneration:** The process by which there is regrowth of new volunteer trees from old stumps under a favourable environment.
- b) Artificial regeneration:** Involves the deliberate rate planting of new forest tree seedlings in deforested areas.

Advantages of Natural Regeneration:

- i. It is less expensive when compared with artificial regeneration.
- ii. It does not require any special management skill.
- iii. It does not require formal stages in plantation establishment.

5. Afforestation

Afforestation is the process of establishing forest plantations in any area. It involves complete removal of natural vegetation before planting new forest species.

Advantages of Afforestation:

- i. Provides a regular supply of raw materials (e.g., timber) for industries.
- ii. Prevents desert encroachment.
- iii. Prevents leaching of plant nutrients.
- iv. Leads to addition of organic matter, resulting in an increase in soil fertility.
- v. Increases soil moisture retention.
- vi. Increases the amount of rainfall in an area.

6. Taungya System

Taungya system is defined as the planting of arable crops at early stages of forest establishment, where the arable crops are harvested before the trees form canopies. It is the integration of agriculture with forestry.

Conditions Necessary for the Practice of Taungya System Include:

- i. Scarcity of land.
- ii. Over-population.
- iii. Unemployment or underemployment of productive youths.
- iv. Where government policies permit the integration of forestry and agriculture.
- v. Low living standard of people.

Advantages of Taungya System / Farming:

- i. A variety of crops are harvested.
- ii. Availability of crop produce throughout the year.
- iii. Where land is scarce, the farmer has a piece of land to cultivate.
- iv. Increases the income of the farmer.
- v. When leguminous crops are used, the beneficial effect of root nodules increases soil fertility for the benefit of the forest trees.

Disadvantages of Taungya Farming:

- i. Monotony of operations:** The practice of taungya farming may be monotonous as the farmer performs the same operation every season.
- ii. Inability of some crops to survive:** Most crops cultivated under taungya farming may not survive due to the invasion of some insects in the forest reserve.
- iii. Competition:** Competition between crop and tree for space, water, nutrients, etc.
- iv. Reluctance of foresters:** The forester may be reluctant in releasing fertile soil for farming purposes.
- v. Crop restrictions:** It makes allowances for the cultivation of selected crops only (only annual and biennial crops are allowed to be planted).

Assignment:

1. Read up Agroforestry practices in Nigeria.

NAME OF SCHOOL: Mater Misericordiae Secondary School, Rumuomasi

WEEK: Five

DATE: 9th October, 2026

CLASS: SS2

SUBJECT: Agricultural science

TOPIC: Agroforestry practices in Nigeria

Meaning of Agroforestry

Agroforestry is a land-use management system in which trees or shrubs are deliberately grown together with crops and/or livestock on the same piece of land, either simultaneously or in a planned sequence, in order to obtain increased and sustainable output from the land through better use of available resources (soil, water, light, and nutrients).

In simple terms, agroforestry combines forestry (tree growing) with agriculture (crop and/or animal production) in an integrated and mutually beneficial manner. It is a deliberate practice that seeks ecological and economic interaction between the different components on the farmland.

Key features of agroforestry:

It involves at least two plant species (one usually a woody perennial such as a tree or shrub) or a combination of woody plants and animals.

The system produces more than one product or service (e.g., food, fodder, fuelwood, timber, shade).

The cycle is usually more than one year.

Even the simplest agroforestry system is more complex, ecologically and economically, than a monoculture system.

Practices of Agroforestry

Agroforestry is guided by certain practices and principles which include:

1. **Alley cropping:** Growing arable crops (e.g., maize, cassava, yam) in the alleys/strips between rows of planted trees or shrubs (e.g., *Leucaena*, *Gliricidia*).
2. **Taungya system:** Farmers are allowed to cultivate food crops among newly planted forest tree seedlings until the tree canopy closes; this helps in forest establishment while providing farmers with food crops.
3. **Woodlot/farm woodlot practice:** Growing trees on a portion of farmland purposely for fuelwood, poles, or timber alongside other farm activities.
4. **Shelterbelts/windbreaks:** Planting rows of trees around farmland or homesteads to reduce wind speed, prevent soil erosion, and protect crops.

5. **Home gardens:** A mixture of trees, shrubs, vegetables, and sometimes small livestock grown around the homestead for family use.
6. **Boundary/live fence planting:** Trees or shrubs planted along farm boundaries to serve as fencing, markers, and windbreaks, while also providing fodder or fruits.
7. **Improved fallow:** Fast-growing trees or shrubs (often nitrogen-fixing) are planted during the fallow period to restore soil fertility faster than natural bush fallow.
8. **Apiculture with trees (Agrosilviapiculture):** Beekeeping is combined with tree planting since trees provide nectar and pollen for bees.

Agroforestry Combinations (Systems/Types)

Agroforestry systems are classified based on the components combined:

1. **Agrisilvicultural system (Crops + Trees):** Combination of agricultural crops with trees, e.g., alley cropping, taungya, home gardens, shelterbelts.
2. **Silvopastoral system (Trees + Pasture/Livestock):** Combination of trees with pasture and livestock grazing, e.g., trees planted on grazing land to provide shade and fodder for animals (cattle, sheep, goats).
3. **Agrosilvopastoral system (Crops + Trees + Livestock):** A combination of arable crops, trees, and livestock on the same land unit, e.g., home gardens involving fruit trees, vegetables, and small livestock like poultry or goats.
4. **Silvofishery/Aquasilviculture (Trees + Fish):** Trees (especially mangroves) are combined with fish farming/aquaculture, common in the coastal/mangrove swamp areas of the Niger Delta.
5. **Apiculture-based systems (Trees + Bees):** Beekeeping combined with tree growing, since flowering trees serve as sources of nectar and pollen.
6. **Multipurpose tree/shrub woodlots:** Trees grown in association with crops or pastures, mainly to serve several purposes such as fuelwood, fodder, timber, and soil improvement.

Tree Species Suitable for Agroforestry Practices in Nigeria

Trees used in agroforestry are usually selected for fast growth, nitrogen-fixing ability, multiple uses, and adaptability to local conditions. They include:

(a) Leguminous/Nitrogen-fixing trees (for soil improvement and fodder):

- i. Gliricidia sepium
- ii. Leucaena leucocephala
- iii. Acacia albida (Faidherbia albida)
- iv. Senna siamea (Cassia siamea)
- v. Calliandra calothyrsus

(b) Fruit trees (for food and income):

- i. Mango (Mangifera indica)
- ii. Orange/citrus (Citrus species)
- iii. Cashew (Anacardium occidentale)

- iv. Guava (*Psidium guajava*)
- v. Pawpaw (*Carica papaya*)

(c) Timber/economic trees:

- i. Teak (*Tectona grandis*)
- ii. Gmelina (*Gmelina arborea*)
- iii. Iroko (*Milicia excelsa*)
- iv. Mahogany (*Khaya* species)

(d) Oil-yielding and multipurpose trees:

- i. Oil palm (*Elaeis guineensis*)
- ii. Shea butter tree (*Vitellaria paradoxa*)
- iii. Neem (*Azadirachta indica*) – also used for pest control and shade

(e) Fodder trees/shrubs (for livestock feed):

- i. *Gliricidia sepium*
- ii. *Leucaena leucocephala*
- iii. Moringa (*Moringa oleifera*)

(f) Mangrove species (for coastal agroforestry/silvofishery):

- i. Red mangrove (*Rhizophora* species)
- ii. White mangrove (*Avicennia* species)

Assignment :

- 1. Define Floriculture.
- 2. List three importance of ornamental plants.

NAME OF SCHOOL: Mater Misericordiae Secondary School, Rumuomasi

WEEK: Six

DATE: 16th October, 2026

CLASS: SS2

SUBJECT: Agricultural science

TOPIC: FLORICULTURE/ ORNAMENTAL PLANTS

Definition of Floriculture

Floriculture is the branch of horticulture concerned with the cultivation, production and marketing of flowers and ornamental plants for gardens, landscaping, indoor decoration and floral art (e.g. bouquets, wreaths). It also involves the growing of plant materials used to beautify homes, offices, streets and public places.

Meaning and Importance of Ornamental Plants

Ornamental plants are plants that are grown mainly for their beauty (flowers, leaves, form or fragrance) rather than for food, fibre or other economic products. They are used to decorate and beautify the environment.

Importance of Ornamental Plants

1. They beautify homes, offices, schools, streets and public places.
2. They provide shade and improve the microclimate around buildings and along roads.
3. They help purify the air by absorbing carbon dioxide and releasing oxygen.
4. They reduce noise and dust pollution when planted as hedges/windbreaks.
5. They provide privacy and serve as boundaries or fences (live hedges).
6. They are a source of income through floriculture and landscaping business.
7. They provide raw materials for the perfume and cosmetic industries.
8. They create a relaxing, attractive environment that improves mental well-being.
9. Some serve cultural, religious and ceremonial purposes (e.g. decoration at weddings, funerals, festivals).

Common Types of Ornamental Plants

- i. Bedding plants — mostly flowering plants grown in beds/borders for colour display (e.g. marigold, zinnia, celosia, cockscomb, petunia).
- ii. Hedging plants — shrubs planted close together and trimmed to form a live fence or boundary (e.g. Duranta, Ixora, Ficus, croton).
- iii. Lawn plants — grasses grown and maintained as a lawn/turf (e.g. Bermuda grass, carpet grass, St. Augustine grass).
- iv. Landscape plants — trees, shrubs and palms used to design and beautify large open spaces, parks and compounds (e.g. royal palm, flamboyant tree, bougainvillea).
- v. Pot plants — ornamentals grown in containers/pots for indoor or patio decoration (e.g. money plant, dracaena, peace lily, aglaonema).

Examples of Ornamental Plants

- i. Rose (*Rosa* species)
- ii. Hibiscus (*Hibiscus rosa-sinensis*)
- iii. Bougainvillea (*Bougainvillea spectabilis*)
- iv. Croton (*Codiaeum variegatum*)
- v. Ixora (*Ixora coccinea*)
- vi. Canna lily (*Canna indica*)
- vii. Marigold (*Tagetes* species)
- viii. Palm trees (e.g. *Roystonea regia* — royal palm)
- ix. Flamboyant/flame tree (*Delonix regia*)
- x. Dracaena and Ficus species

Setting and Locations for Ornamental Plants

a. Identification of Locations/Settings Requiring Ornamental Plants

- i. Avenues / Streets — roadside trees, flowering shrubs and hedges.
- ii. Building environment — plants around offices, homes and schools (pot plants, foundation shrubs, climbers).
- iii. Gardens — flower beds, lawns and landscaped features in parks and private/public gardens.

b. Determination of Ornamental Plant for Each Location

The choice of plant for a location depends on factors such as available space, soil type, amount of sunlight, purpose (shade, screening, beauty), climate, and maintenance requirements. For example, tall trees such as royal palms suit wide avenues, while low flowering bedding plants suit garden borders, and compact pot plants suit indoor spaces.

c. Recognition of Ornamental Plants Using Common and Botanical Names

Every ornamental plant has a common name (used locally) and a botanical/scientific name (Latin binomial, internationally recognized). Students should be able to match examples, such as:

- i. Rose — *Rosa* species
- ii. Hibiscus — *Hibiscus rosa-sinensis*
- iii. Bougainvillea — *Bougainvillea spectabilis*
- iv. Flame tree/Flamboyant — *Delonix regia*
- v. Royal palm — *Roystonea regia*

Assignment: Visit any garden, school compound or avenue near you. List at least three ornamental plants you find there, state their common and botanical names, classify each under its type (bedding, hedging, lawn, landscape or pot plant), and describe one maintenance operation being carried out on them.